

Functional Nutrition

Introduction and Overview



KELECHI A. UDUHIRI, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosure

Kelechi A. Uduhiri, MD disclosed no relevant financial relationships with any ineligible companies.

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Functional Medicine Certification Competencies

I. PATIENT EVALUATION

A) PHYSICAL SIGNS AND PHYSICAL EXAM

- i. Identify evidence of disturbances in function and clinical imbalances in the patient's physical exam for signs of nutrient deficiency, insufficiency, excess, and toxicity, inclusive of anthropometric data, hygiene, mood and affect.

B) LABORATORY AND OTHER EVALUATIVE TESTING

- i. Determine if further quantitative data is necessary to provide additional predictive, prognostic, and/or diagnostic specificity after completion of the qualitative patient evaluation.
- ii. In these scenarios, perform, interpret, or refer for evaluative tests using analytes and biomarkers capable of providing clinical evidence of disturbances in physiological, structural, mental, or emotional function, and that may improve predictive, prognostic, and/or diagnostic specificity.

Functional Medicine Certification Competencies

II. THERAPEUTIC INTERVENTION

- a) Evaluate patient readiness to engage in dietary, lifestyle, and behavioral changes prior to initiating therapeutic interventions; prioritize these as initial interventional modalities.
- b) According to their ability to favorably modulate function, correct clinical imbalances, and/or address each patient's personalized root causes, and in alignment with the patient's goals, values, preferences, sociocultural context, and what gives them meaning and purpose:
 - i. Identify, prioritize, select, and initiate the following therapeutic interventions when clinically indicated:
 1. Dietary and other lifestyle strategies.
 2. Behavioral change strategies.
 3. Pharmaceuticals, nutraceuticals, botanicals, and other supplements.
 4. Mind-body and other modalities.
 - ii. Monitor changes in function in response to therapeutic interventions and adjust regimen accordingly.
 - iii. Refer and/or collaborate with additional healthcare practitioners, as needed or required, to align with the patient's goals and desired outcomes.

Clinical Guidelines

NIH: Dietary Guidelines for Americans 2020-2025

<https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials>

USPSTF: Screening for Food Insecurity

https://jamanetwork.com/journals/jama/fullarticle/2831307?guestAccessKey=37a3cec2-db5f-4289-8041-74181ffe3977&utm_source=silverchair&utm_medium=email&utm_campaign=article_alert-jama&utm_content=etoc&utm_term=041525&utm_adv=003501890927

Clinical Guidelines

NIH: Nutritional Assessment: StatPearls

<https://www.ncbi.nlm.nih.gov/books/NBK580496/>

Position of the Academy of Nutrition and Dietetics: Malnutrition (Undernutrition) Screening Tools for All Adults

[https://www.jandonline.org/article/S2212-2672\(19\)31366-8/pdf](https://www.jandonline.org/article/S2212-2672(19)31366-8/pdf)

Role of the Nutrition-focused Physical Examination in Identifying Malnutrition and Its Effectiveness

<https://aspenjournals.onlinelibrary.wiley.com/doi/10.1002/ncp.10797>

Academy of Nutrition and Dietetics/American Society for Parenteral and Enteral Nutrition Consensus Malnutrition Characteristics: Usability and Association With Outcomes

<https://pubmed.ncbi.nlm.nih.gov/31074906/>

Clinical Guidelines

ASPEN: Nutrition-Focused Physical Assessment for Micronutrient Deficiencies

<https://aspenjournals.onlinelibrary.wiley.com/doi/10.1177/0884533615573054>

AND: Nutrition-Focused Physical Exam

<https://www.eatright.org/health/health-conditions/malnutrition-and-deficiencies/what-is-the-nutrition-focused-physical-exam>

Consensus Statement: Academy of Nutrition and Dietetics and American Society for Parenteral and Enteral Nutrition: Characteristics Recommended for the Identification and Documentation of Adult Malnutrition (Undernutrition)

<https://pubmed.ncbi.nlm.nih.gov/22535923/>

Position of the Academy of Nutrition and Dietetics: Micronutrient Supplementation

<https://pubmed.ncbi.nlm.nih.gov/30366569/>

Food and Agriculture Organization of the United Nations: Global Dietary Guidelines

<https://www.fao.org/nutrition/nutrition-education/food-dietary-guidelines/en/>

Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain the functional nutrition tenet: “Food is energy, information, community, and medicine.”
2. Describe the scope of nutrient deficiencies at the population level.
3. List foundational tools and methods of patient evaluation in the functional nutrition model.

Clinical Focus

- Foundational principles of functional nutrition.



FUNCTIONAL NUTRITION

In the Functional Medicine and Whole Health Model

Remove what causes imbalance.

Provide what creates balance.

Whole Systems and Personalization

Functional Nutrition

- Focuses on high-quality, nutrient-dense foods
- Focuses on phytonutrient diversity
- Addresses clinical imbalances to move individuals toward the highest expression of health



IFM's Food Plan Suites

- Educate patients in healthy eating
- Provide options to personalize nutrition plans and/or therapeutic diets





Functional Nutrition Fundamentals

Eating Your Way to Better Health

Functional nutrition is about finding the right way for each of us to eat—using food to maximize the potential for health and reverse dysfunction or disease. There is no single “right diet” that applies to everyone. We have different genetic backgrounds, different preferences, and different lives. We all want to be healthy, but many of us haven’t figured out exactly how to make food and dietary patterns serve that goal. Functional nutrition offers the concepts, strategies, and tools to make that happen.

Functional Nutrition Fundamentals

For untold centuries, humans have relied on the food supply as a source of energy, health, and connection. However, over the last several decades, changes in the food supply (and in how we use it) have contributed strongly to the growing epidemic of chronic disease. Functional nutrition developed out of a desire on the part of healthcare providers to change that picture.

Emerging science is very clear that food is a powerful influence on health. Food offers not only the calories that fuel our body’s metabolism (engine), but it also contains many diverse components that play important roles in all our bodily functions. Poor-quality food can create disease, and high-quality food (in the right proportions and amounts) can reverse disease and sustain health. In a very real sense, food is **medicine**!

How Food Affects Our Health

There are three basic elements in foods:

- **Macronutrients:** the proteins, fats, and carbohydrates that are the building blocks of all food
- **Micronutrients:** vitamins and minerals
- **Phytonutrients:** substances in plants that have a positive effect on health

The abundance of processed foods has negatively impacted our nutrition and health. Food manufacturing practices like stripping whole grains of their fiber, adding sweeteners and excessive salt to foods, and applying toxic chemicals to agriculture can have a negative impact on our health. When processed foods are eaten regularly, they can make us sick.

Food is **information**, not just energy or calories. Food tells our bodies how to function by signaling when and how each type of cell should behave in every situation. So, we want the very best information to reach our cells to help achieve optimal functioning.

Food also represents **connection**. Food connects us when we prepare and eat together and enjoy celebrations or ceremonies, affirming cultural bonds.

How Does Functional Nutrition Work?

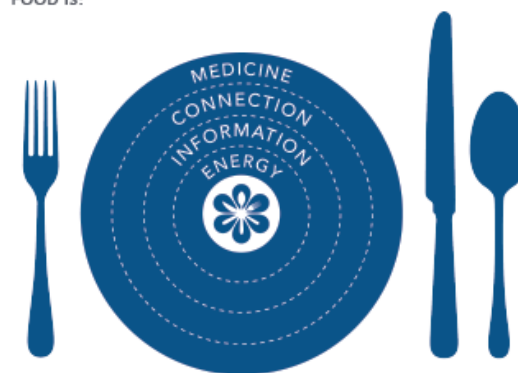
Functional nutrition emphasizes healthful eating personalized to your genetics, lifestyle, environment, and health concerns. Functional nutrition offers strategies and tools to improve your health, including recipes, cooking, and food storage recommendations, plus steps to achieve mindful eating.

When you work with a functional nutrition practitioner to help you understand and implement your personalized food plan, you will have access to a wealth of information to help you adopt a new approach to food and nutrition. Your provider will also determine if other interventions are needed.

Changing eating behaviors isn’t easy, even if you are well-informed about what to do. There will be ups and downs. But, unlike a diet, these changes are not harsh restrictions on your eating, and they do not involve counting calories or depriving yourself of delicious food. They are scientifically sound approaches that can benefit you for a lifetime! Just start with the following basic concepts: **eat fresh, eat color, eat texture**, and (from bestselling author Michael Pollan) **eat real food, not too much, mostly plants**.

Welcome to functional nutrition. It is the best way to harness the energy, information, and connections of food to create health and reverse disease!

FOOD IS:

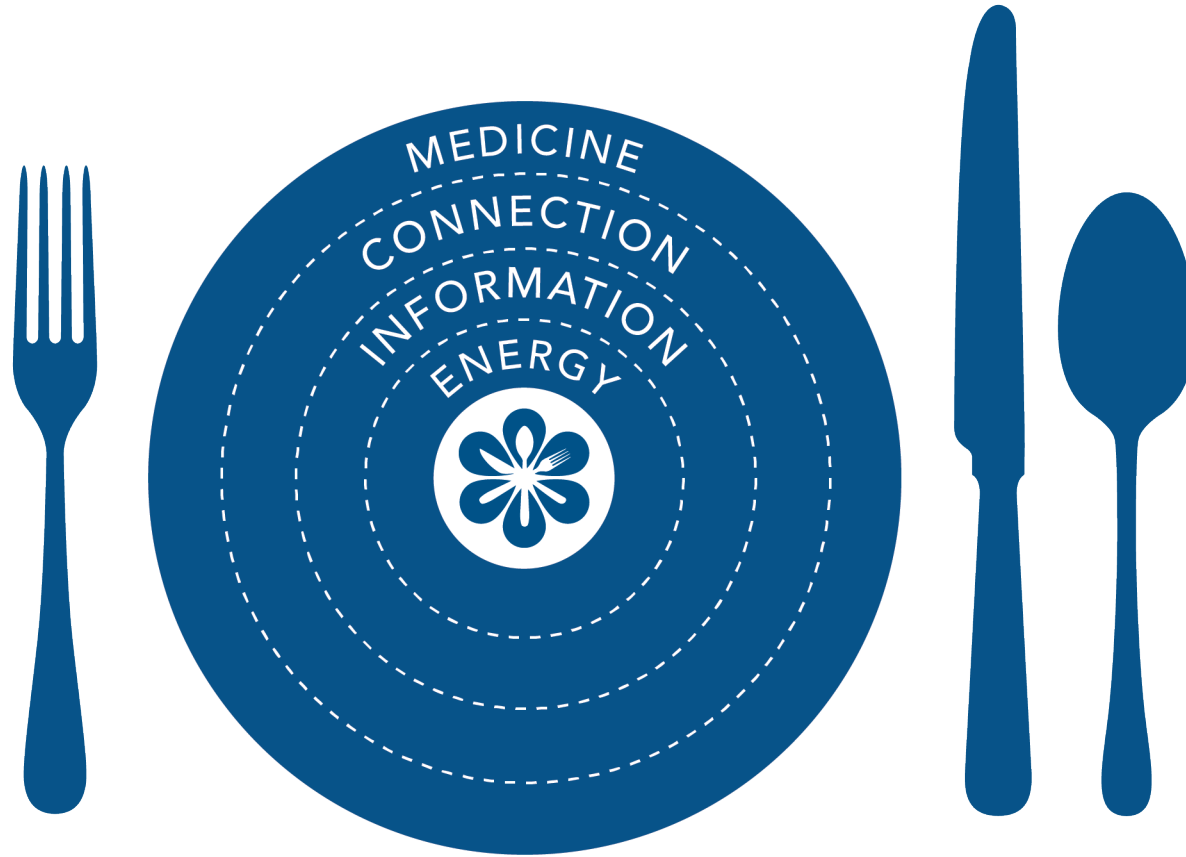


Clinical Focus

- The multidimensional functions of food.



Food Is...

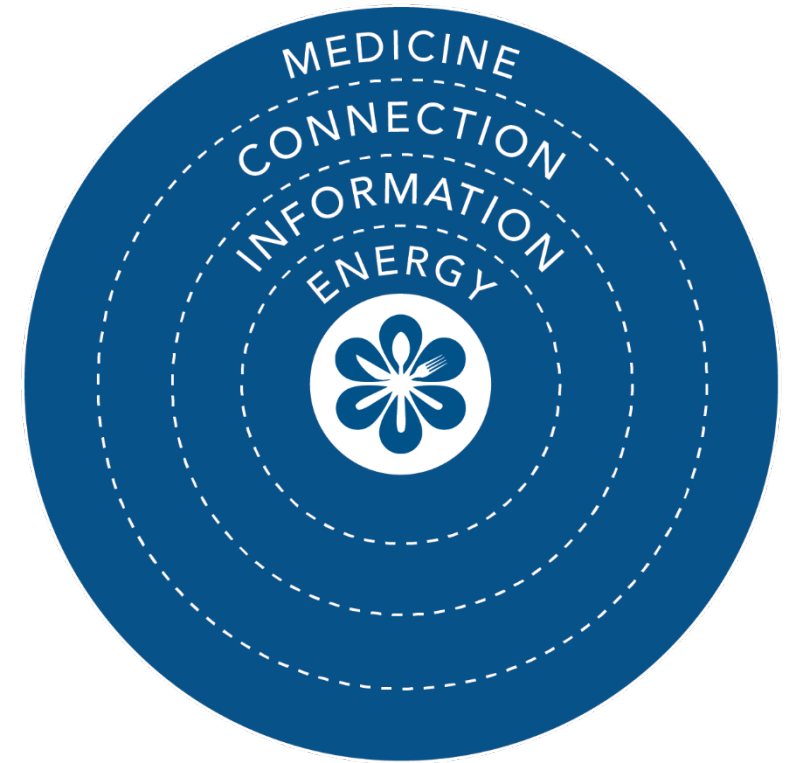


Food is energy

Macronutrients and Micronutrients

Cellular Fuel

Catalysts & Cofactors

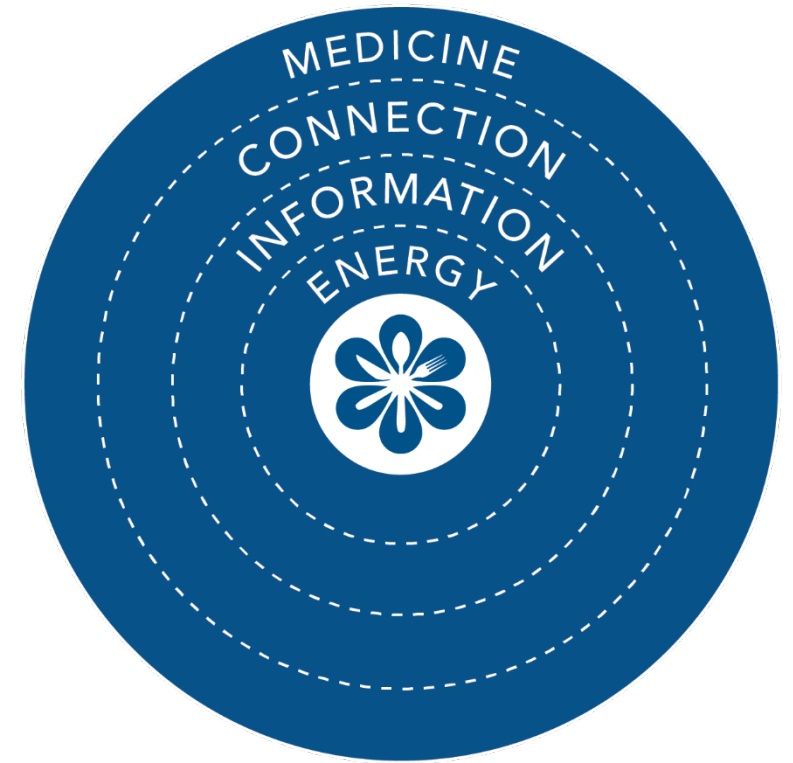


Food is information

We are not our genes.

We are the expression of our genes.

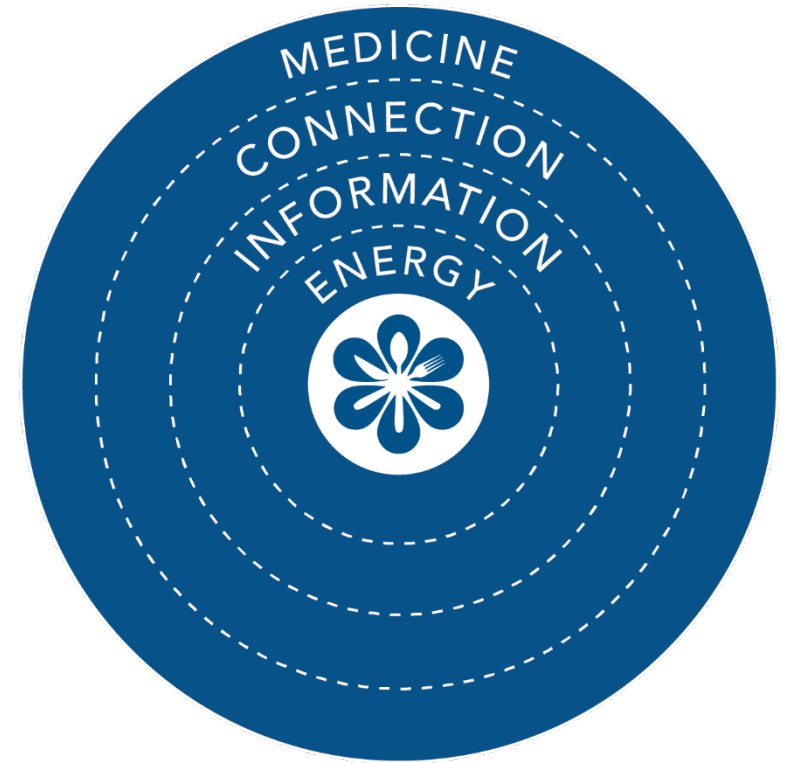
Foods modulate gene expression.



Food is connection

We are who our friends are.

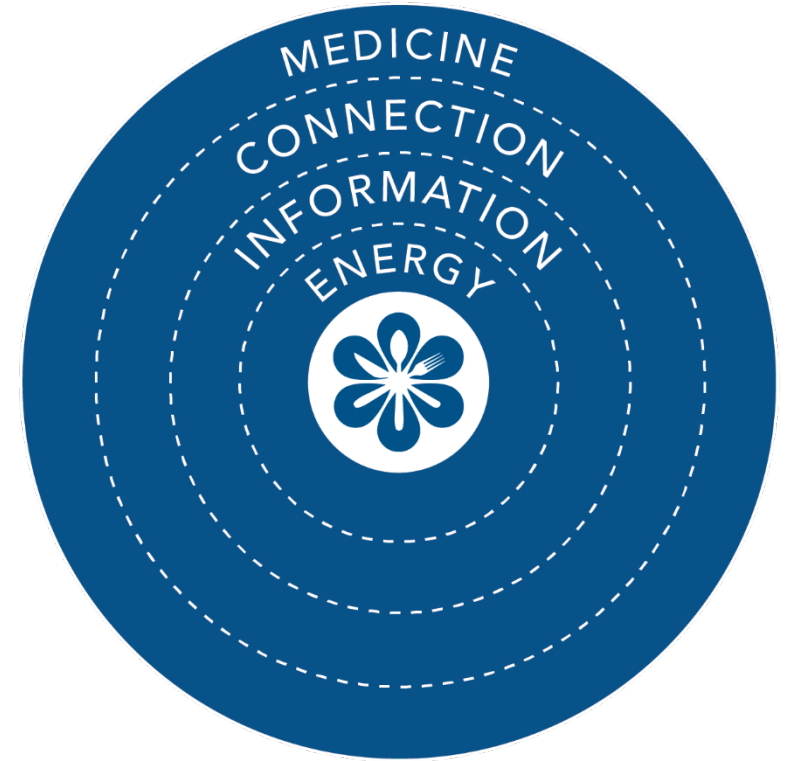
We are what our friends eat.



Food is medicine

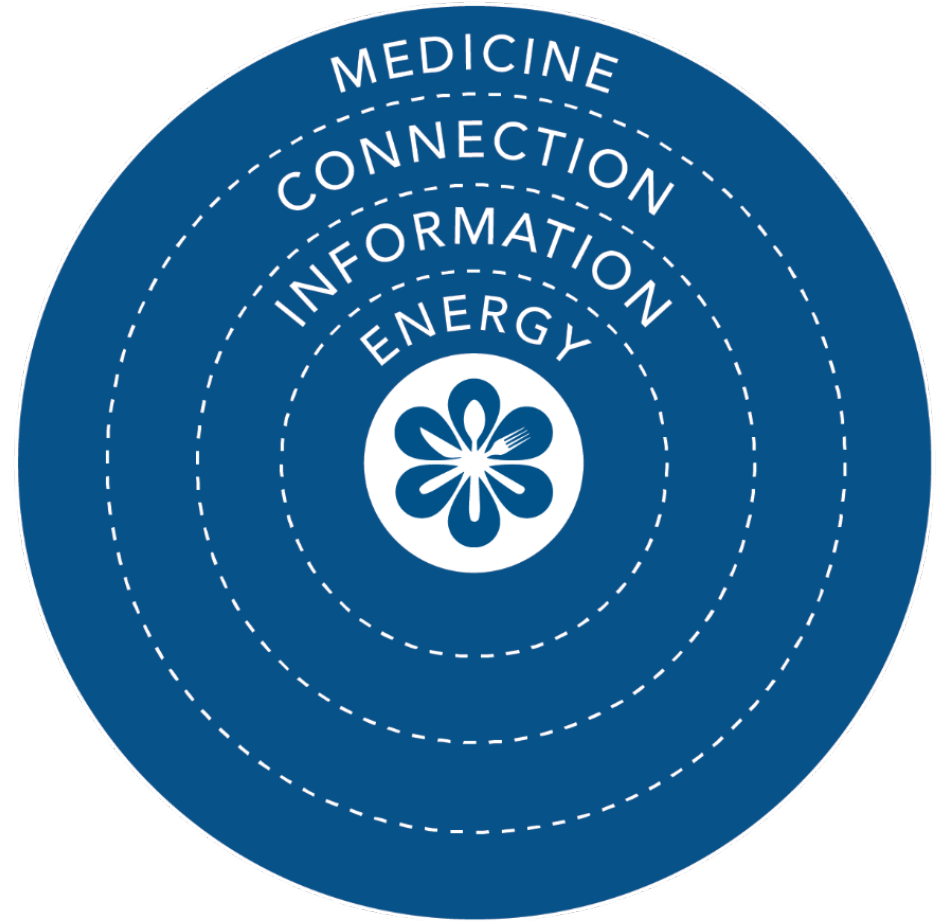
“Let food be your medicine and medicine be your food.”

- Hippocrates



We are...

- What we eat
- Where we eat
- When we eat
- How much we eat
- With whom we eat
- Why we eat
- Why we don't eat



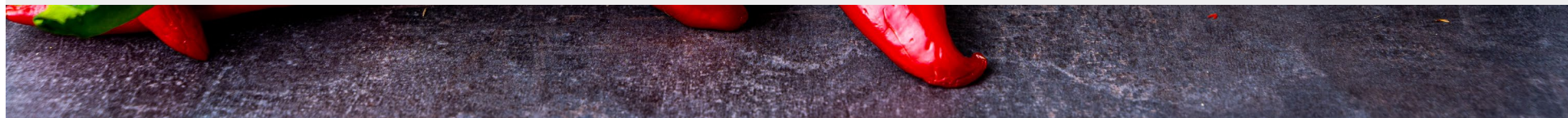
Clinical Focus

- Breadth and prevalence of micronutrient deficiencies.





**What Are You Eating?
Are You Getting Enough Nutrients to Function Well?**



Percent of Adult Population (Ages 19-71) Below Estimated Average Requirement

Nutrients	Food Only	Food + Supplement
Calcium	27-72	22-47
Copper	4.1-9.6	3.1-6.7
Iron	<1-6.2	<1-5.0
Magnesium	47.8-68.6	41.9-55.2
Zinc	10.1-26.1	8-16.4
Vitamin A	37-47	23-38
Thiamin	4.2-8.9	3.5-5.3
Riboflavin	2.5-3.6	1.9-2.1
Folate	8.3-17	6.4-10.3
B6	4.0-22.4	3.2-14
B12	2.3-5.2	1.7-3.3
Vitamin C	25-44	25-35
Vitamin D	94.6-95.5	44.4-73.5
Vitamin E	83.8-91.7	52.1-64.1

UK NUTRITION SURVEY (Age 20-50) – PERCENT BELOW LOWER REFERENCE NUTRIENT INTAKE (LRNI)

The amount of a nutrient that is enough for only a small number of people who have lower requirements.

Nutrients	% Male <LRNI	LRNI	% Female <LRNI	LRNI
Vitamin A (ug)	11.3	300	6.75	250
Riboflavin (mg)	3.8	0.8	12.6	0.8
Folate (ug)	1.8	100	4.4	100
B12 (ug)	1.15	1	2.1	1
Selenium (ug)	25.8	40	50.3	40
Potassium (mg)	10.0	2,000	24.3	2,000
Magnesium (mg)	14.2	190	11.5	150
Iron (mg)	1.1	8.0	25.2	4.7
Zinc (mg)	8.2	5.5	5.2	4.0
Iodine (ug)	5.7	70	11.3	70

Clinical Focus

- Breadth and diversity of populations affected by micronutrient deficiencies.

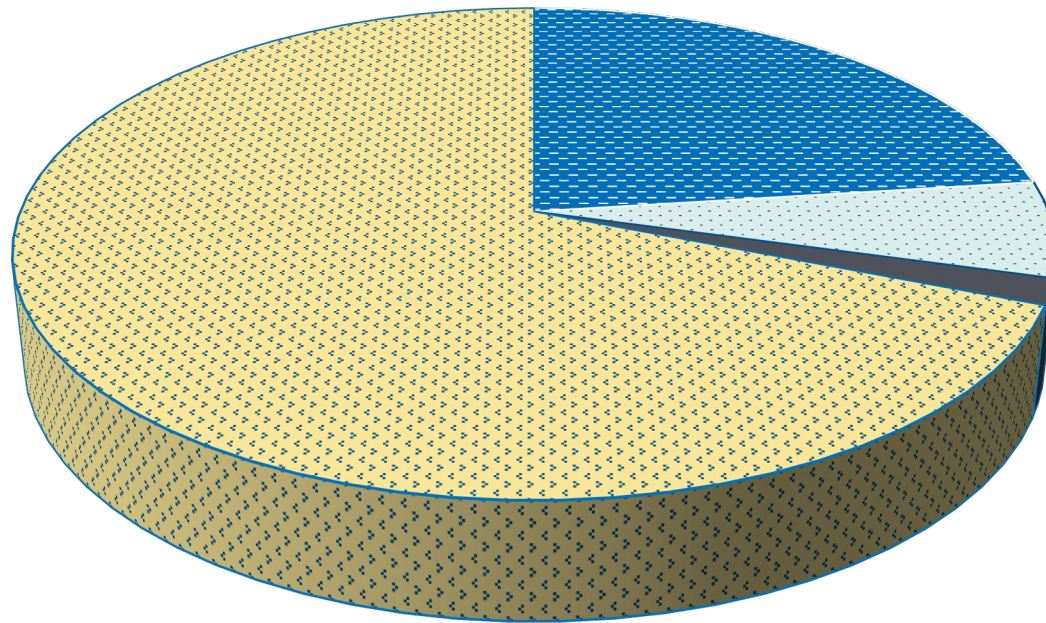


THE CHALLENGE

Recognize nutrition insufficiency in a society of “abundance.”

NHANES USA – Risk of Deficiency In Micronutrients (n>15,000)

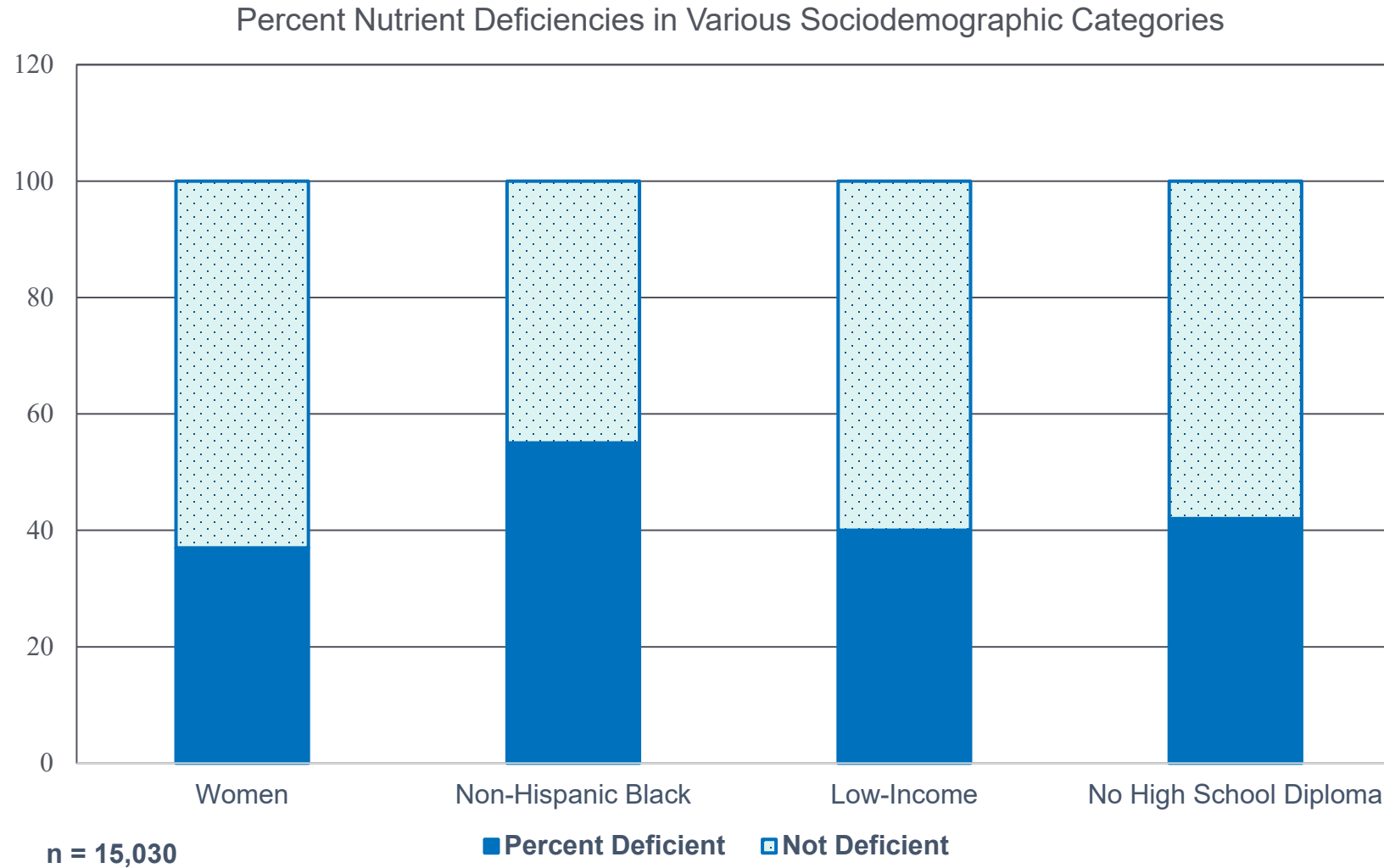
% Population with nutrient deficiency



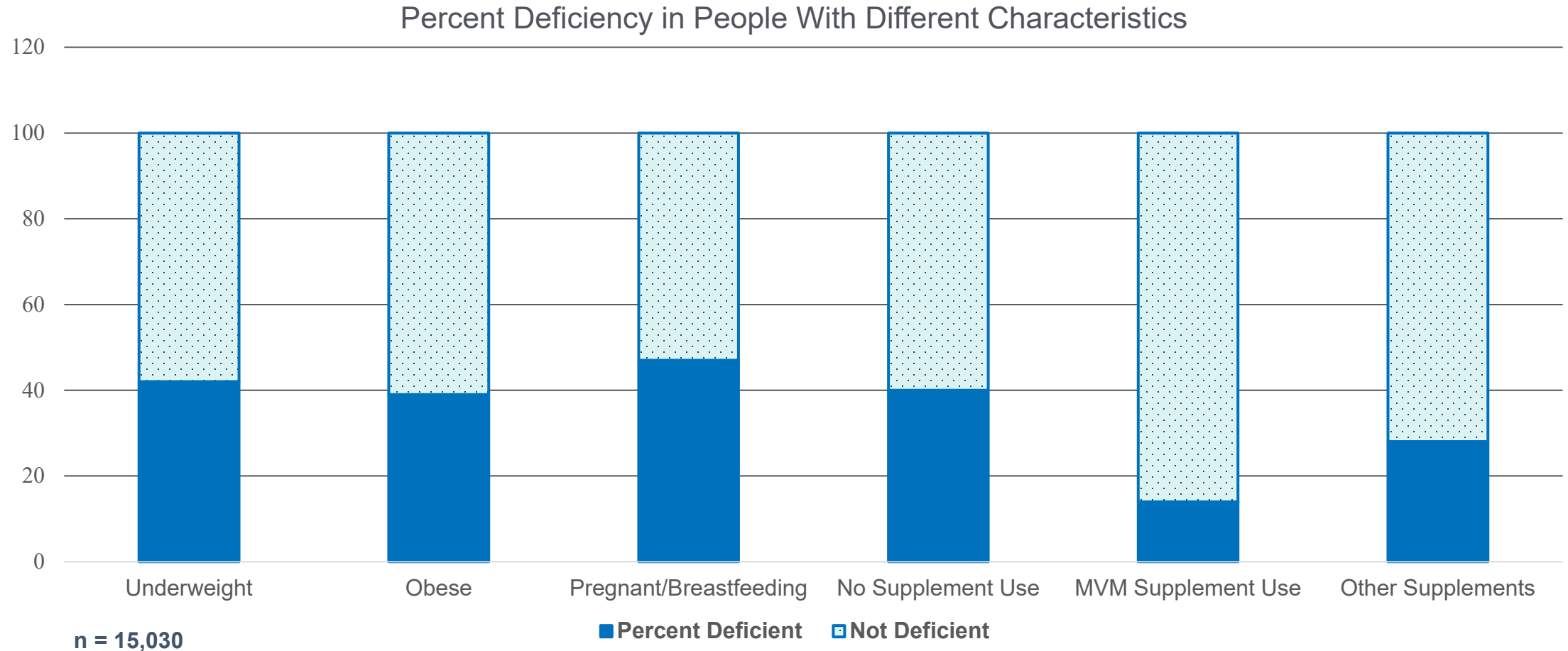
■ 1 nutrient ■ 2 nutrients ■ 3-5 nutrients ■ no nutrient deficiency

- 23% have 1 nutrient deficiency
- 6.3% have 2 nutrient deficiencies
- 1.7% have 3-5 nutrient deficiencies

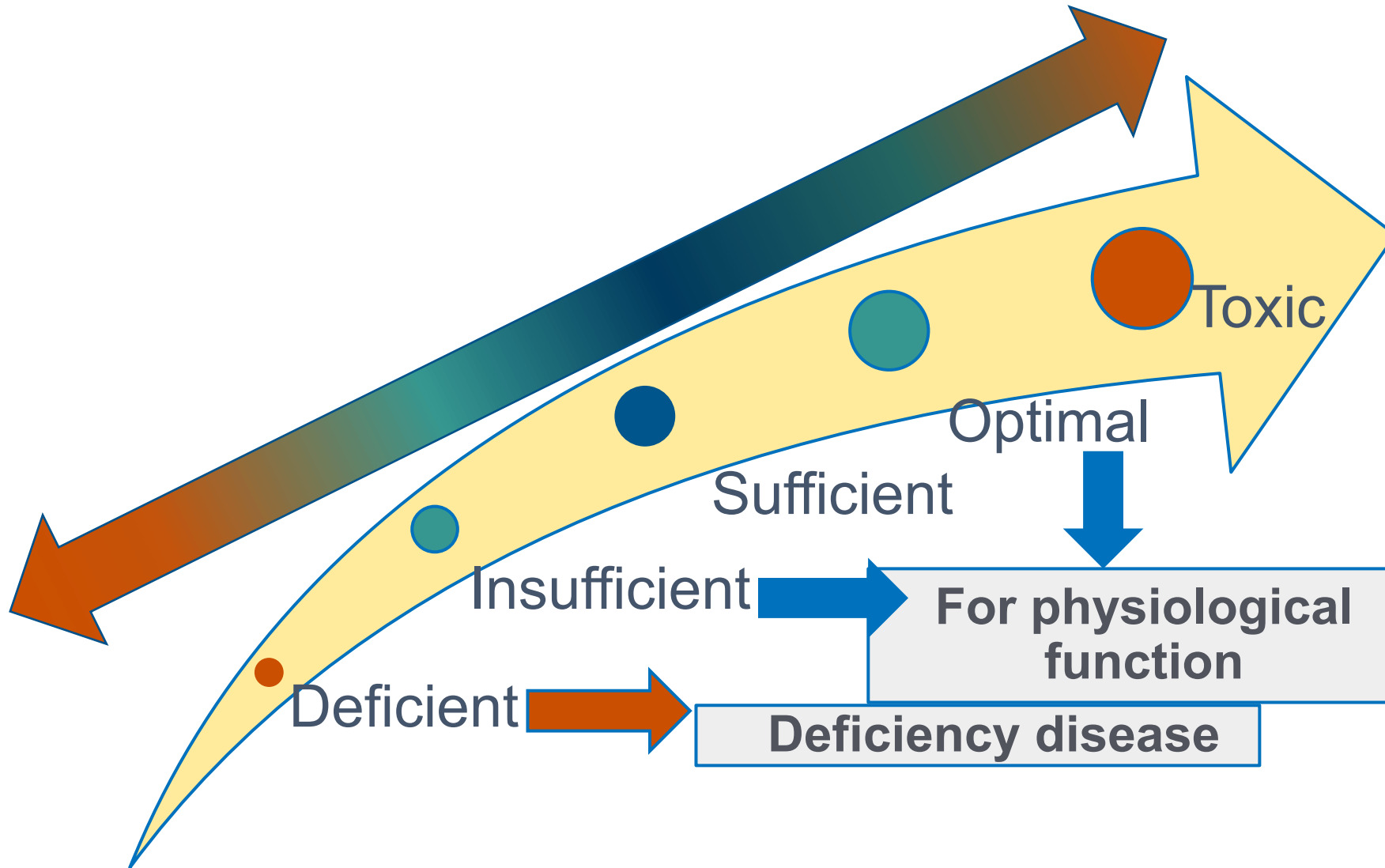
NHANES USA – Risk of Deficiency in Micronutrients



NHANES USA – Risk of Deficiency in Micronutrients



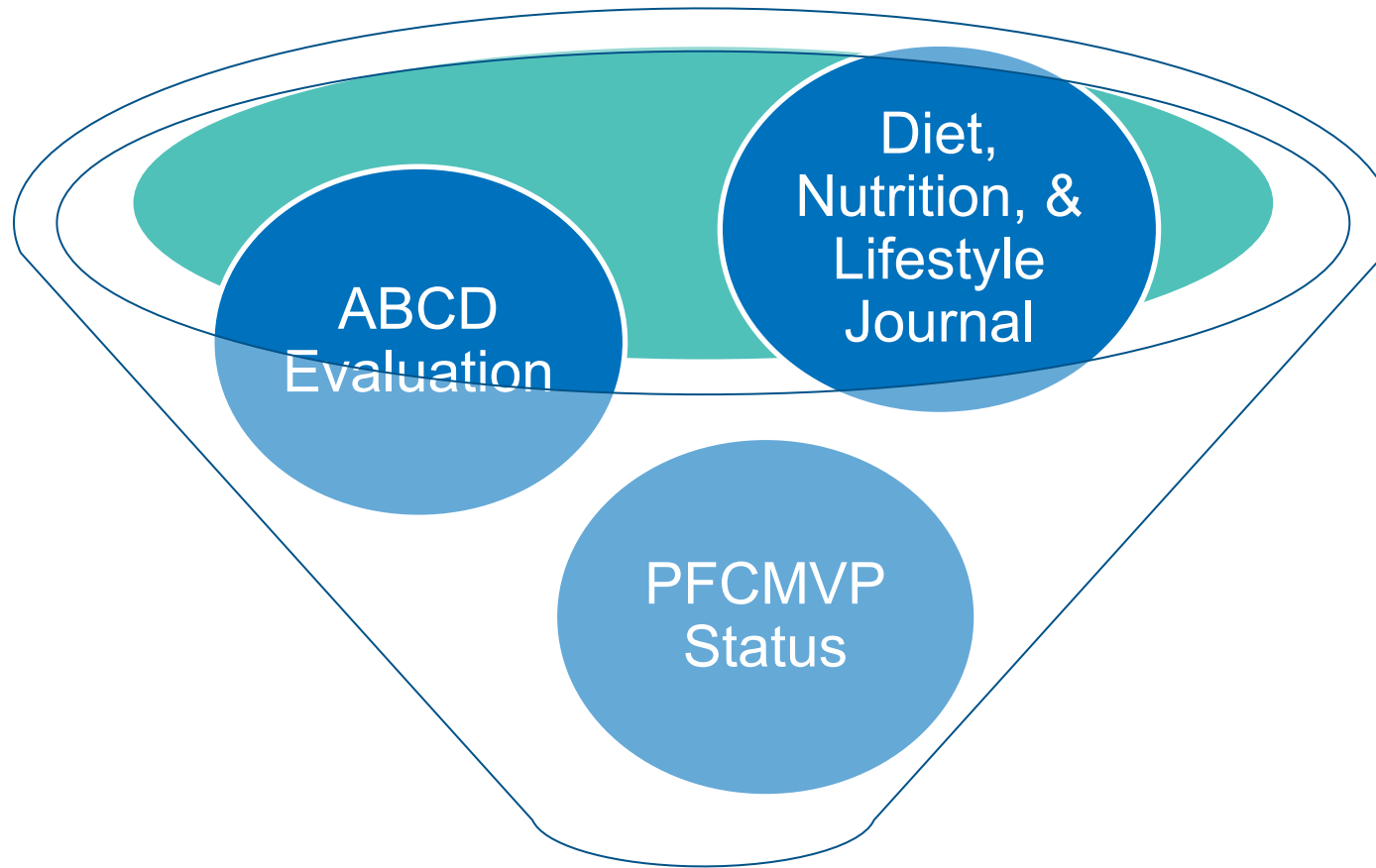
CONTINUUM OF NUTRIENT NEEDS



Clinical Focus

- Patient assessment tools and methods.





Gathering



Diet, Nutrition & Lifestyle Journal—1 Day

Patient Name _____

Date _____

Food Plan Type _____

Day 1

Day Event	Food & Drink Intake (include type, amount, brand)	Macronutrients (PFC) and Phytonutrients
Rising Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Breakfast Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Mid-AM Snack Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Lunch Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Mid-PM Snack Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Dinner Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
PM Snack Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR
Bed Time		P _____ F _____ C _____ ☐ R ☐ O ☐ Y ☐ G ☐ B/P/BL ☐ W/T/BR

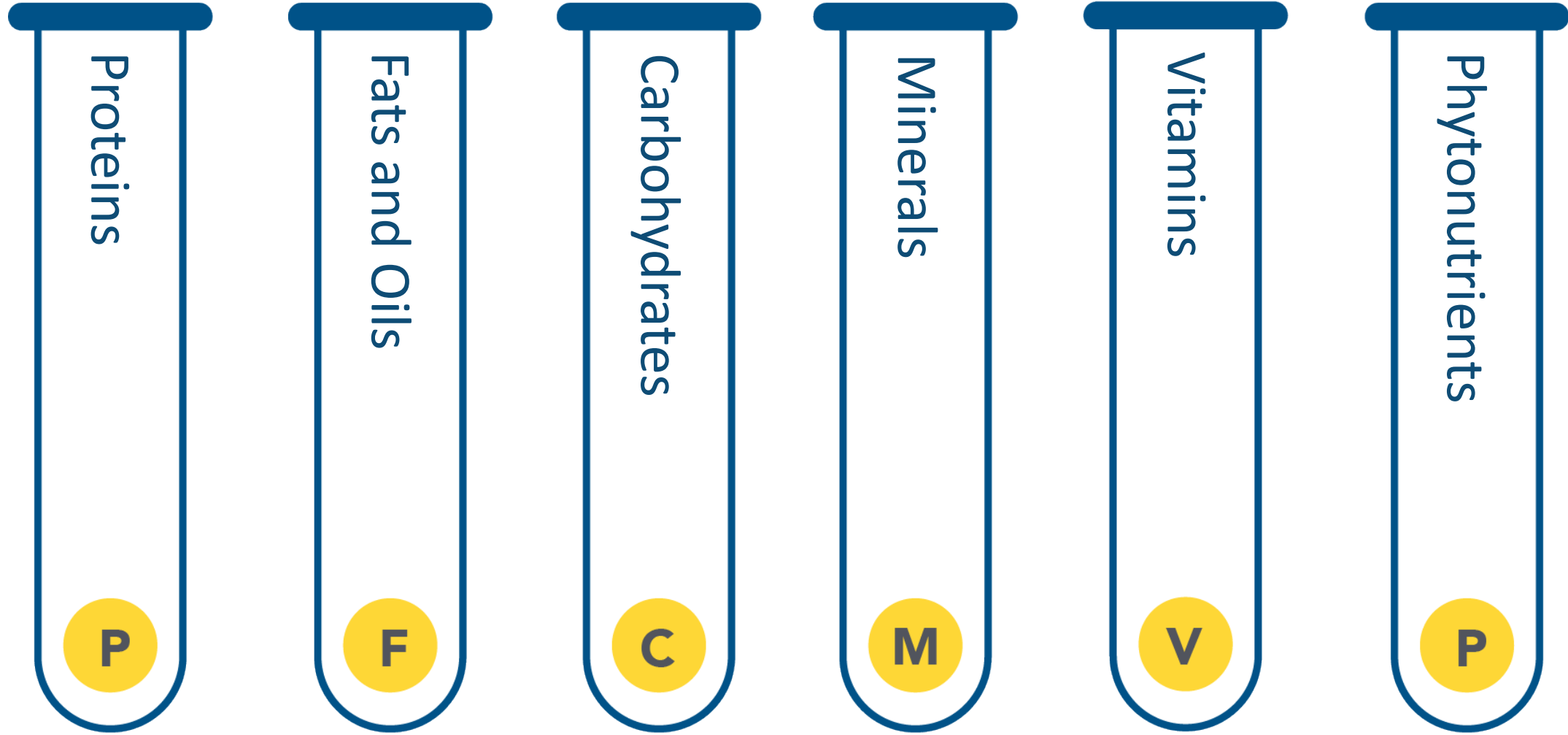
P: Proteins; F: Fats; C: Carbohydrates; R: Red; O: Orange; Y: Yellow; G: Green; B/P/BL: Blue/Purple/Black; W/T/BR: White/Tan/Brown

Sleep & Relaxation	Exercise & Movement	Stress	Relationships
Sleep Quantity _____ (hrs) Quality: ☐ Poor ☐ Fair ☐ Good Relaxation ☐ Yes ☐ No Type/Amount:	Type, Duration, & Intensity ☐ Aerobic: ☐ Strength: ☐ Flexibility:	Stress reduction practices: Stressors:	Supporting: Non-supporting:

Mental	Emotional	Spiritual

The PFCMVP Mnemonic

Six critical categories of nutrients



ABCDs of the Functional Nutrition Evaluation



Anthropometrics
& Vital Signs



Biomarkers &
Functional Labs



Clinical Findings
(Nutrition Exam)



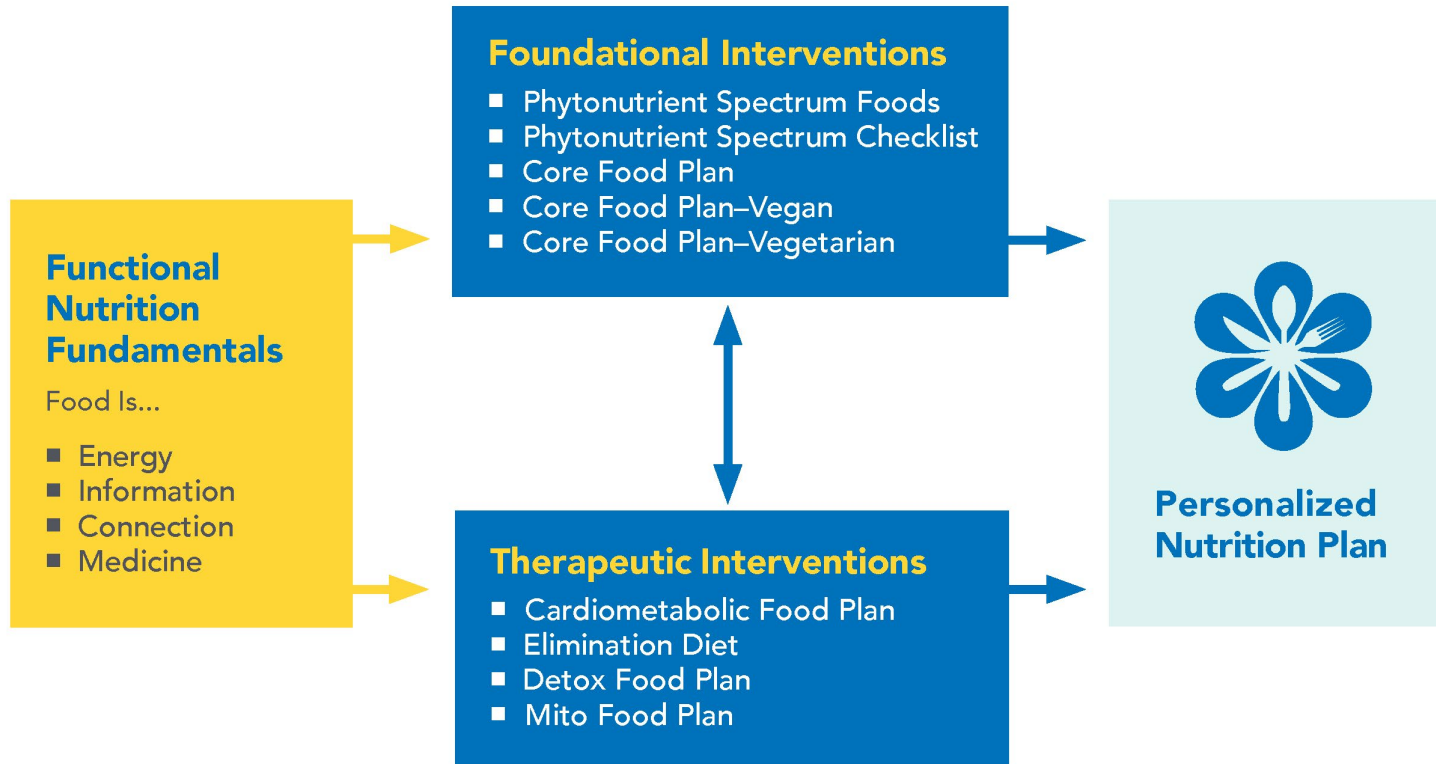
Diet and Lifestyle
Assessment

The Suite of IFM Therapeutic Food Plans

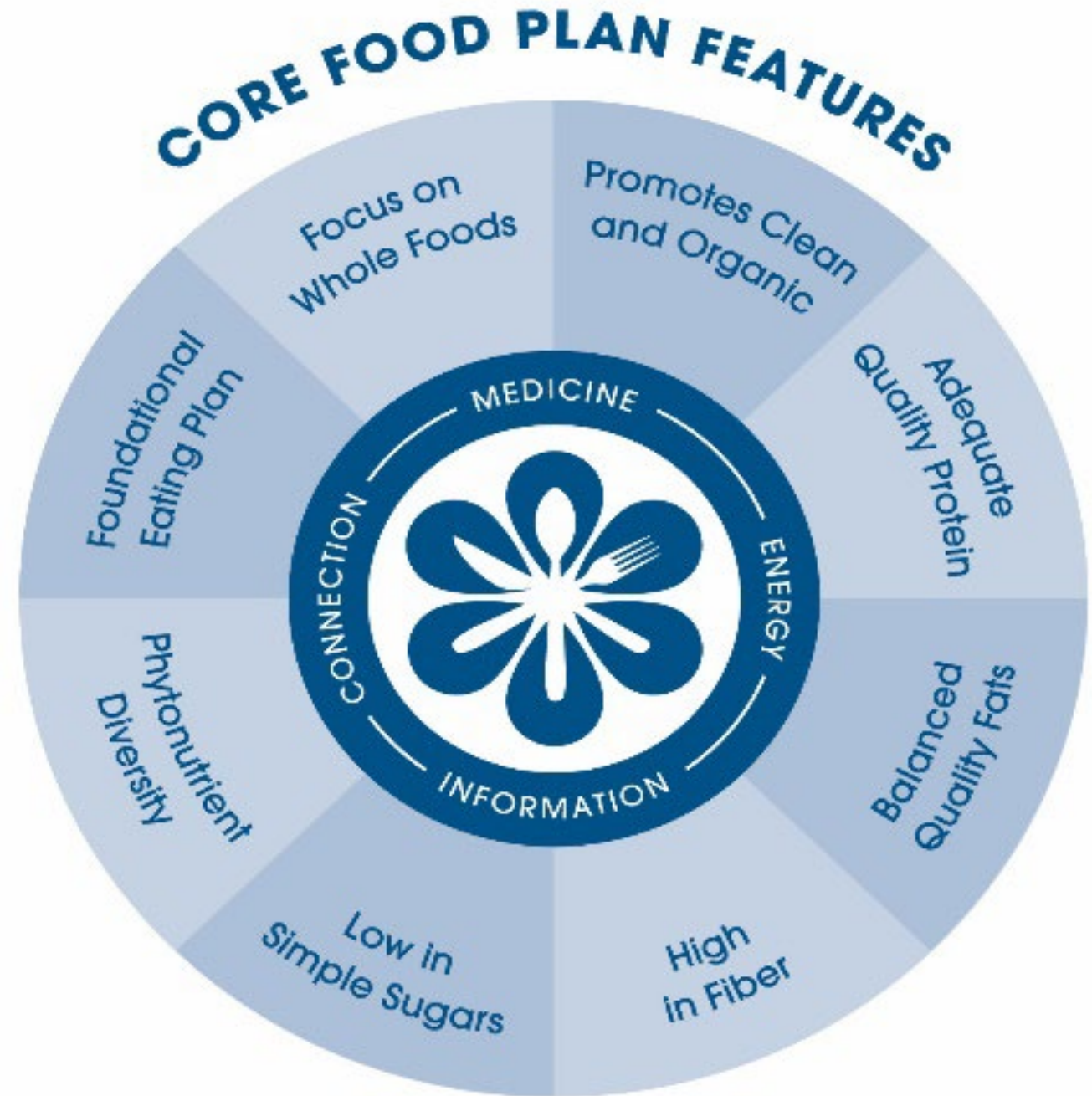
Phyto – Core – Cardio – Elimination – Detox – Mito



Nutrition Interventions

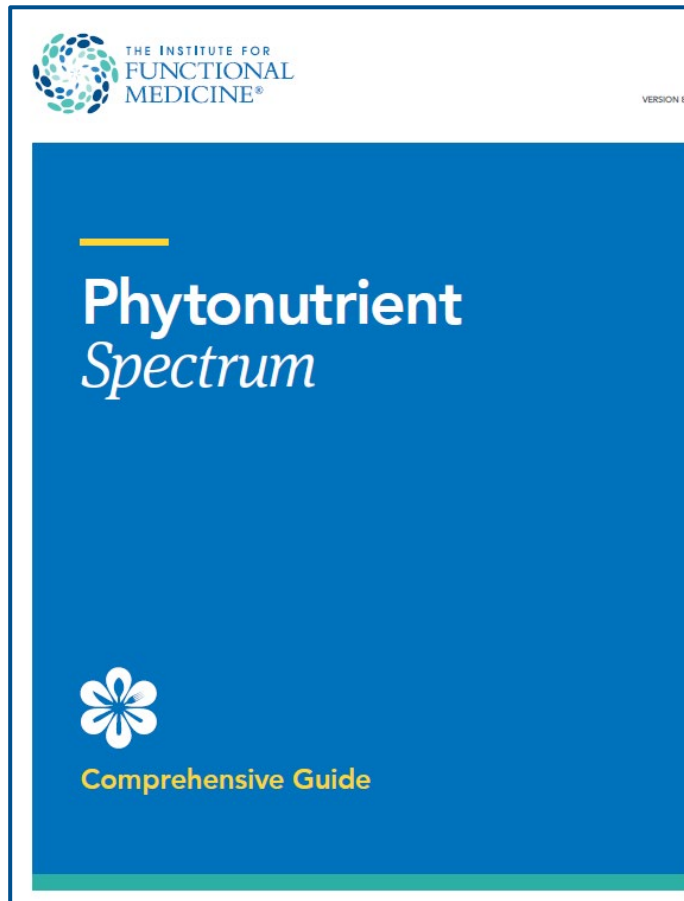


Therapeutic Food Plans



Phytonutrient Spectrum

PATIENT EDUCATION RESOURCES



Phytonutrient Spectrum Foods

Red

Primary phytonutrients in red foods: Anthocyanins and anthocyanidins,^{1,2} betalains,^{1,3} and carotenoids such as lycopene.¹

Foods Apples, red skinned Beans, red (adzuki, kidney, etc.) Beets Bell pepper, red Blood oranges Cranberries Cherries Grapefruit, pink Goji berries Grapes, red skinned	Guava Onions, red Plums Pomegranate Potatoes, red Prickly pear Radicchio Radishes Raspberries Strawberries Sweet peppers, red	Rhubarb Rooibos tea Tomato Watermelon
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Eating red foods and their unique phytonutrients is associated with:

- Reduced inflammation^{4,7}
- Reduced total and LDL cholesterol^{8,9}
- Reduced blood pressure¹⁰⁻¹⁷
- A lower risk of cardiovascular disease^{18,19}
- Improvements in memory and cognitive function²⁰
- A reduced risk of lung, head and neck, prostate, and breast cancers²¹⁻²⁴
- Increased beneficial gut bacteria²⁵⁻²⁸
- Increased protection of the skin from sun damage²⁹
- Protection against cell damage^{30,31}

Orange

Primary phytonutrients in orange foods: Carotenoids, including alpha-carotene, beta-carotene, beta-cryptoxanthin, lutein, and zeaxanthin.^{32,33}

Foods Apricots Bell peppers, orange Cantaloupe Carrots Mango Nectarine Orange	Papaya Persimmons Pumpkin Squash (acorn, butternut, winter) Tangerines Turmeric root	Sweet potato (sometimes referred to as "yams" in the U.S.) Squash (acorn, butternut, winter) Tangerines Turmeric root
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Eating orange foods and their unique phytonutrients is associated with:

- A decreased risk of type 2 diabetes³⁴
- A lower risk of coronary heart disease (CHD)^{35,36}
- A reduced risk of breast, cervical, head, neck, and bladder cancers^{37,38,39}
- A lower risk of osteoporosis and bone fractures⁴⁰⁻⁴²
- A reduced risk of asthma^{43,44}
- A healthy sperm count^{45,46}

Yellow

Primary phytonutrients in yellow foods: Primary phytonutrients in yellow foods: Carotenoids, such as alpha-carotene, lutein and zeaxanthin.¹

Foods Apples, yellow skinned Asian pears Banana Bell peppers, yellow Corn Ginger root	Gold kiwi Jackfruit Lemon Millet Olive oil, extra virgin Passionfruit	Pineapple Plantains Potatoes, yellow Starfruit Summer squash
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Eating yellow foods and their unique phytonutrients is associated with:

- A lower risk of asthma^{47,48}
- A reduced risk of metabolic syndrome and type 2 diabetes^{49,50}
- A decreased risk of stroke⁵¹
- Healthy vision⁵²
- A lower risk of breast, head, neck, bladder and prostate cancers^{53,54,55,56}
- Protection against cell damage^{57,58}

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Green

Primary phytonutrients in green foods: Chlorophyll, catechins, glucosinolates, and carotenoids.⁴⁴

Foods Apples, green skinned Artichoke Asparagus Avocado Bamboo sprouts Bean sprouts Bell peppers, green Bok choy Broccoli Broccoli Brussels sprouts Cabbage, green Celery Chayote	Cucumbers Edamame/Soybeans Feijoa Green beans Green peas Green tea Greens (beet, chard, collard, dandelion, kale, lettuce, mustard, spinach, turnip) Herbs (basil, cilantro, parsley, thyme) Kiwi	Limes Nopales Okra Olive oil, extra virgin Olives, green Pears, green skinned Peppers, green Snow peas Tomatillos Watercress Zucchini
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Eating green foods and their unique phytonutrients is associated with:

- A reduced risk of lung, bladder, colon, ovarian, pancreatic, stomach, prostate, and kidney cancers^{59,60}
- A lower risk of depression⁶¹
- A decreased risk of coronary heart disease (CHD) and stroke⁶²
- Healthy brain function with aging^{63,64}
- Protection against cell damage⁶⁵⁻⁶⁸
- Enhanced detoxification of harmful chemicals^{69,70}
- Healthy estrogen detoxification^{71,72}

Blue/Purple/Black

Primary phytonutrients in blue/purple/black foods: Anthocyanins/anthocyanidins¹ and resveratrol³

Foods Bell pepper, purple Berries (blueberries, blackberries, boysenberries, huckleberries, marionberries) Cabbage, purple Carrots, purple Cauliflower, purple	Eggplant Figs Grapes, blue/purple/black skinned Kale, purple Olives, purple/black Plums Potatoes, purple Prunes	Raisins Rice, purple/black
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Eating blue/purple/black foods and their unique phytonutrients is associated with:

- Maintenance of healthy cognitive function^{73,74}
- Decreased blood pressure^{75,76}
- A reduced risk of cardiovascular disease¹⁰
- Protection from liver damage caused by inflammation^{77,78}
- Reduced fasting blood sugar and hemoglobin A1c⁷⁹
- Less muscle soreness and faster recovery after exercise⁸⁰
- Healthy levels of immune cells that defend the body against harmful microbes^{81,82}

White/Tan/Brown

Primary phytonutrients in white/tan/brown foods: Sulfur compounds, such as allicin; flavones;¹ and lignans.⁷⁹

Foods Apples Appleauce Cassava root (yuca root) Cauliflower Charimeya Cocoa Coconut Coffee Dates Garlic Jicama	Legumes (chickpeas, dried beans or peas, hummus, lentils, peanuts) and legume dips Lychee Mushrooms Nuts (almonds, cashews, pecans, walnuts) Onions Pears Pitaya (dragon fruit)	Seeds (flax, hemp, pumpkin, sesame, sunflower) Shallots Soy Tahini Taro root Tea (black, white) Turnips Whole grains (amaranth, barley, brown rice, oat, quinoa, rye, spelt, teff, wheat)
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Eating white/tan/brown foods and their unique phytonutrients is associated with:

- A decreased risk of breast, colon, esophageal, stomach, and pancreatic cancers⁸³⁻⁸⁷
- Decreased total and LDL cholesterol^{88,89}
- A reduced risk of cardiovascular disease and healthier, more elastic blood vessels⁹⁰⁻⁹³
- A decreased risk of type 2 diabetes^{94,95}

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